

UC SAN DIEGO / ECE 251C

1

Filters and LTI systems

Impulse responses, frequency response, and linear phase

How does an impulse response determine a filter?

What distinguishes a finite impulse response (FIR) filter from an infinite impulse response (IIR) filter? What is an impulse response in the first place? We begin with linearity and time invariance, then use a moving average to connect filter coefficients with frequency response. Symmetry will lead us to the four types of linear-phase FIR filters.

1.1 Signals, systems, and linearity

A discrete-time signal is a sequence $x = (x[n])_{n \in \mathbb{Z}}$, where $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$. The symbol $x[n]$ is one sample; x denotes the entire sequence. A system H maps an input sequence x to an output sequence y :

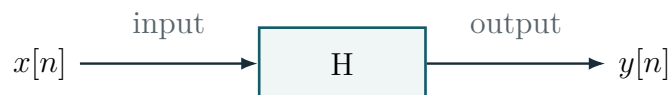


Figure 1.1: The system acts on the whole sequence: $y = Hx$.

What does it mean for a system to be linear? If $y_1 = Hx_1$ and $y_2 = Hx_2$, then linearity means

$$H(\alpha x_1 + \beta x_2) = \alpha y_1 + \beta y_2 \quad (\alpha, \beta \in \mathbb{R}). \quad (1.1)$$

Scaling and adding inputs has exactly the same effect as scaling and adding the corresponding outputs. In particular, a linear system sends the zero input to the zero output.

1.2 Time invariance and the impulse response

Linearity concerns *amplitudes*. Time invariance concerns *location*. A time-invariant system applies the same rule no matter where the input is placed on the time axis.

For an integer k , define the shift T_k by

$$(T_k x)[n] := x[n - k].$$

When $k > 0$, this delays the signal by k samples. If $y = Hx$, time invariance requires $H(T_k x) = T_k y$. Equivalently,

$$HT_k = T_k H \quad (k \in \mathbb{Z}). \quad (1.2)$$

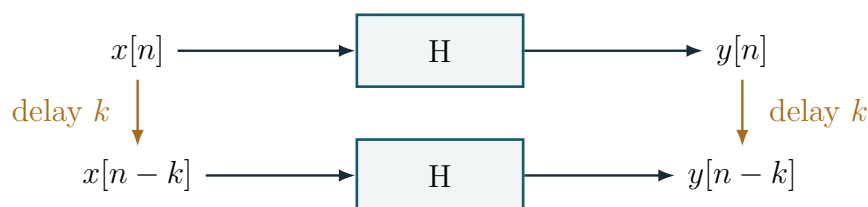


Figure 1.2: Delaying the input delays the output by the same amount. The two routes from the upper-left signal to the lower-right signal agree.

We call a system *linear time invariant*, or *LTI*, when it has both properties. The informal statement that it “does not care when it is turned on” means precisely the shift identity (1.2); we are comparing shifted whole input sequences, not changing an internal initial state.

What is an impulse response? The unit impulse, or Kronecker delta, is the sequence

$$\delta[n] := \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$

The *impulse response* is the output $h := H\delta$ when the input is δ .

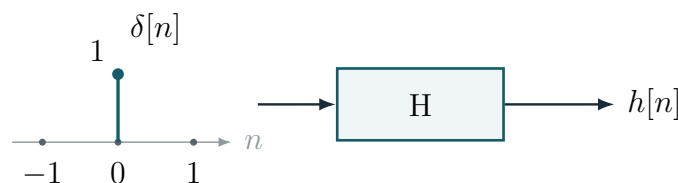


Figure 1.3: One unit sample at time zero probes the system. Its output is the impulse response.

Time invariance sends $\delta[n - k]$ to $h[n - k]$, and linearity sends $a\delta[n - k]$ to $ah[n - k]$. We therefore know the response to every scaled and shifted impulse.

1.3 Why an LTI system is described by convolution

Every finitely supported sequence can be assembled from impulses:

$$x[n] = \sum_{k \in \mathbb{Z}} x[k] \delta[n - k]. \quad (1.3)$$

For a fixed n , the only surviving term has $k = n$. Thus the coefficient of the impulse at location k is exactly the sample $x[k]$.

Apply the system to this finite linear combination. First use linearity, and then time invariance:

$$\begin{aligned} y[n] &= \left[H \left(\sum_k x[k] T_k \delta \right) \right] [n] \\ &= \sum_k x[k] (H T_k \delta)[n] \\ &= \sum_k x[k] (T_k H \delta)[n] \\ &= \sum_k x[k] h[n - k]. \end{aligned} \quad (1.4)$$

Changing the summation index to $r = n - k$ gives

$$y[n] = (h * x)[n] := \sum_{r \in \mathbb{Z}} h[r] x[n - r]. \quad (1.5)$$

This is *convolution*. The system response is a superposition of shifted copies of h , each weighted by an input sample.

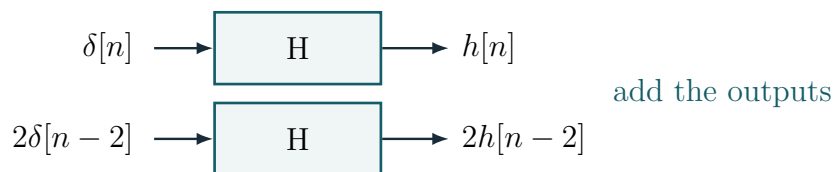


Figure 1.4: For $x[n] = \delta[n] + 2\delta[n - 2]$, the output is $y[n] = h[n] + 2h[n - 2]$. The same construction works for any finite collection of impulses.

Where convergence matters. The derivation above uses a finite sum of input impulses. Extending that derivation to infinite-duration inputs requires the system to respect the relevant limit of finite sums. For an FIR filter, the right side of (1.5) has finitely many terms and defines the convolution system for every input sequence. For infinite filters, a sufficient condition for pointwise absolute convergence is $\sum_r |h[r]| < \infty$ and bounded x , since

$$\sum_r |h[r] x[n - r]| \leq \left(\sup_m |x[m]| \right) \sum_r |h[r]| < \infty.$$

1.4 FIR, IIR, and the moving average

The impulse response h is also called the *filter*. Its *support* is the set of indices where its coefficients are nonzero:

$$\text{supp}(h) := \{n \in \mathbb{Z} : h[n] \neq 0\}.$$

A filter is *finite impulse response* (FIR) if $|\text{supp}(h)| < \infty$, and *infinite impulse response* (IIR) if $|\text{supp}(h)| = \infty$. Here $|\text{supp}(h)|$ counts the elements of the support; it does not sum the coefficient magnitudes.

The distinction concerns the response to *one impulse*, not the duration of the input. An FIR filter may process an input that lasts forever, but every output sample depends on only finitely many input samples.

Example: what does this system do? Consider

$$y[n] = \frac{1}{2}(x[n] + x[n-1]). \quad (1.6)$$

It averages two neighboring samples. Substituting $x = \delta$ gives

$$h[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1], \quad \text{supp}(h) = \{0, 1\}.$$

Thus it is a two-tap FIR filter. Conversely, substituting these two nonzero coefficients into (1.5) gives exactly (1.6).

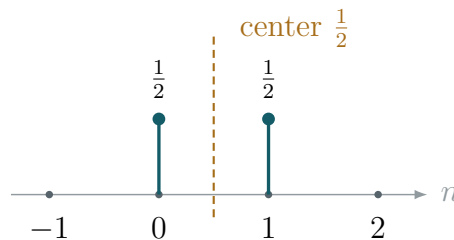


Figure 1.5: The moving-average impulse response. Two equal coefficients lie on opposite sides of a half-integer center. This symmetry will reappear in its phase response.

Two inputs make its smoothing effect explicit. If $x[n] = c$ is constant, then $y[n] = (c + c)/2 = c$. If $x[n] = (-1)^n$ alternates at every sample, then

$$y[n] = \frac{1}{2}((-1)^n + (-1)^{n-1}) = 0.$$

It preserves a constant signal and removes the fastest alternating pattern. These checks suggest a *lowpass* filter. To describe what happens at the frequencies between these extremes, we need its frequency response.

1.5 Pure frequencies and the frequency response

Write $j^2 = -1$. Euler's identity $e^{j\theta} = \cos \theta + j \sin \theta$ combines a cosine and sine into one convenient input. We extend a real linear filter to complex sequences by applying it separately to the real and imaginary parts.

For a pure frequency $x[n] = e^{j\omega n}$, convolution gives

$$\begin{aligned} (h * x)[n] &= \sum_{k \in \mathbb{Z}} h[k] e^{j\omega(n-k)} \\ &= e^{j\omega n} \sum_{k \in \mathbb{Z}} h[k] e^{-j\omega k} \\ &= H(e^{j\omega}) e^{j\omega n}, \end{aligned} \tag{1.7}$$

where

$$H(e^{j\omega}) := \sum_{k \in \mathbb{Z}} h[k] e^{-j\omega k}. \tag{1.8}$$

The output has the *same frequency*; the filter only multiplies it by a complex number. In the language of linear algebra, complex exponentials are eigenfunctions of the convolution system, with eigenvalue $H(e^{j\omega})$.



Figure 1.6: An LTI filter does not create new frequencies from a single exponential. Its frequency response specifies the gain and phase at each frequency.

If $H(e^{j\omega}) = \rho e^{j\theta}$ with $\rho \geq 0$, then the output is $\rho e^{j(\omega n + \theta)}$. For a real filter, taking real parts yields

$$\cos(\omega n) \longmapsto \rho \cos(\omega n + \theta).$$

Thus $|H(e^{j\omega})|$ is the magnitude response, and an argument of $H(e^{j\omega})$ is the phase response when $H \neq 0$.

For an FIR filter, this calculation involves only a finite sum. It also holds when $\sum_k |h[k]| < \infty$, since $|e^{j\omega(n-k)}| = 1$ and the series is absolutely convergent.

1.6 Convolution becomes multiplication

For an absolutely summable sequence x , its *discrete-time Fourier transform* (DTFT) is

$$X(e^{j\omega}) := \sum_{n \in \mathbb{Z}} x[n] e^{-j\omega n}.$$

Equation (1.8) is the same transform applied to h . Both transforms are 2π -periodic: replacing ω by $\omega + 2\pi$ multiplies every summand by $e^{-j2\pi n} = 1$.

Proposition 1.1 (Convolution theorem). If $\sum_n |x[n]| < \infty$ and $\sum_n |h[n]| < \infty$, then $y = h * x$ is absolutely summable and

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}). \quad (1.9)$$

Proof. The bound

$$\sum_n \sum_k |h[k]x[n-k]| = \sum_k |h[k]| \sum_m |x[m]| < \infty$$

shows both that y is absolutely summable and that we can reorder the double series below. Substitute $m = n - k$:

$$\begin{aligned} Y(e^{j\omega}) &= \sum_n \left(\sum_k h[k]x[n-k] \right) e^{-j\omega n} \\ &= \sum_k h[k] \sum_m x[m] e^{-j\omega(m+k)} \\ &= \left(\sum_k h[k] e^{-j\omega k} \right) \left(\sum_m x[m] e^{-j\omega m} \right). \end{aligned}$$

The two factors are H and X . □

The z -transform. Define

$$X(z) := \sum_{n \in \mathbb{Z}} x[n] z^{-n}, \quad z \in \mathbb{C} \setminus \{0\}, \quad (1.10)$$

The z -transform is defined where the series converges. On the unit circle, whenever this substitution is valid,

$$X(e^{j\omega}) = X(z)|_{z=e^{j\omega}}.$$

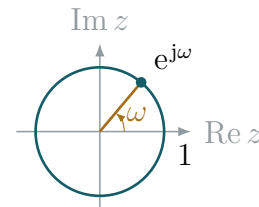


Figure 1.7: The frequency response is the z -transform of the filter evaluated on the unit circle.

1.7 The moving average in the frequency domain

For $h[n] = \frac{1}{2}(\delta[n] + \delta[n - 1])$, the z -transform is

$$H(z) = \frac{1}{2}(1 + z^{-1}).$$

Evaluate on the unit circle and factor around the midpoint of the two powers:

$$\begin{aligned} H(e^{j\omega}) &= \frac{1}{2} + \frac{1}{2}e^{-j\omega} \\ &= e^{-j\omega/2} \frac{e^{j\omega/2} + e^{-j\omega/2}}{2} \\ &= \underbrace{e^{-j\omega/2}}_{\text{phase factor}} \underbrace{\cos(\omega/2)}_{\text{real amplitude}}. \end{aligned} \quad (1.11)$$

The last step is Euler's cosine identity. The exponential factor has magnitude one, so

$$|H(e^{j\omega})| = |\cos(\omega/2)|.$$

On $-\pi < \omega < \pi$, the cosine is positive. A continuous phase choice on this interval is therefore $\varphi(\omega) = -\omega/2$. At $\omega = \pm\pi$ the response vanishes, and its phase is undefined.



Figure 1.8: The moving average preserves zero frequency and suppresses frequencies near $\pm\pi$. Its phase has constant slope $-1/2$. Open endpoints on the phase plot indicate zeros of the response.

The lowpass interpretation is now quantitative. At $\omega = 0$, the gain is one. At $\omega = \pi$, it is zero. At an intermediate frequency such as $\omega = \pi/2$, it is $\cos(\pi/4) = 1/\sqrt{2}$. These agree with our constant-input and alternating-input calculations.

Why does the number $1/2$ occur in the phase? The two coefficients are centered at $L = 1/2$. Factoring out the exponential associated with that center leaves two terms at equal and opposite offsets, which combine into a cosine. We now develop that calculation for a general symmetric filter.

1.8 Linear phase and a center of symmetry

For a real filter, we use the linear-phase factorization

$$H(e^{j\omega}) = e^{j(-L\omega + \beta)} H_{\text{ampl}}(\omega), \quad H_{\text{ampl}}(\omega) \in \mathbb{R}. \quad (1.12)$$

The phase factor is affine in frequency: its slope is $-L$ and its offset is β . In DSP this is called *linear phase*, including the constant offset.

Real amplitude is not necessarily magnitude

The real function H_{ampl} may change sign. The magnitude response is $|H_{\text{ampl}}|$. Where $H_{\text{ampl}} < 0$, the actual phase has an additional π modulo 2π ; at zeros it is undefined. Thus (1.12) does not assert that a principal-value phase plot is one unbroken line.

Consider an FIR filter whose first and last nonzero coefficients occur at 0 and $N - 1$. Its *length* is N , and its center is

$$L := \frac{N - 1}{2}.$$

Length includes any zero coefficients between the endpoints. For example, a five-sample antisymmetric filter can have a zero middle coefficient and still have length five.

Symmetry and antisymmetry about L mean, respectively,

$$h[n] = h[2L - n], \quad h[n] = -h[2L - n]. \quad (1.13)$$

The index $2L - n$ is the reflection of n about L . For a pair of locations $L - r$ and $L + r$, factor out the center exponential:

$$\begin{aligned} h[L - r]e^{-j(L-r)\omega} + h[L + r]e^{-j(L+r)\omega} \\ = e^{-jL\omega} (h[L - r]e^{jr\omega} + h[L + r]e^{-jr\omega}). \end{aligned}$$

Equal coefficients give $2h[L - r] \cos(r\omega)$. An opposite pair gives $2jh[L - r] \sin(r\omega)$. This one pairing identity is responsible for all four types:

	N odd, $L \in \mathbb{Z}$	N even, $L - \frac{1}{2} \in \mathbb{Z}$
symmetric	Type I	Type II
antisymmetric	Type III	Type IV

1.9 Type I: odd length, symmetric

Here $N = 2L + 1$ with L an integer. There is a coefficient at the center, and the other coefficients occur in L reflected pairs.

$$\begin{aligned} H(e^{j\omega}) &= h[L]e^{-jL\omega} + \sum_{r=1}^L h[L-r](e^{-j(L-r)\omega} + e^{-j(L+r)\omega}) \\ &= e^{-jL\omega} \left(h[L] + 2 \sum_{r=1}^L h[L-r] \cos(r\omega) \right). \end{aligned} \quad (1.14)$$

Thus H_{ampl} is a sum of integer-frequency cosines, and $\beta = 0$.

Example. Take

$$(h[0], \dots, h[4]) = \left(\frac{1}{4}, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{4}\right), \quad L = 2.$$

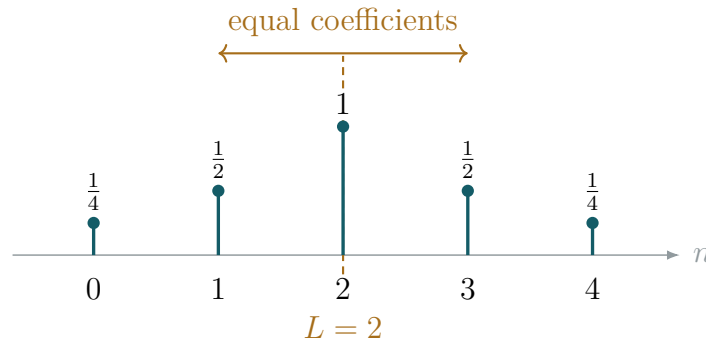


Figure 1.9: An integer center leaves one unpaired coefficient. The two neighboring pairs contribute cosines at offsets one and two.

The frequency response is

$$\begin{aligned} H(e^{j\omega}) &= \frac{1}{4} + \frac{1}{2}e^{-j\omega} + e^{-2j\omega} + \frac{1}{2}e^{-3j\omega} + \frac{1}{4}e^{-4j\omega} \\ &= e^{-2j\omega} \left[1 + \frac{1}{2}(e^{j\omega} + e^{-j\omega}) + \frac{1}{4}(e^{2j\omega} + e^{-2j\omega}) \right] \\ &= e^{-2j\omega} \left[1 + \cos \omega + \frac{1}{2} \cos(2\omega) \right]. \end{aligned}$$

Express the cosine coefficients in terms of h .

In the notation $H_{\text{ampl}}(\omega) = \sum_{r=0}^L a[r] \cos(r\omega)$, equation (1.14) gives

$$a[0] = h[L], \quad a[r] = 2h[L-r] \quad (1 \leq r \leq L).$$

The center is counted once; every reflected pair is counted twice. That distinction explains the different formula for $a[0]$.

1.10 Type II: even length, symmetric

Now $L - \frac{1}{2}$ is an integer. There is no sample at the center. The offsets from the center are $r + \frac{1}{2}$, where $r = 0, \dots, L - \frac{1}{2}$.

Pairing the two equal coefficients at each offset gives

$$\begin{aligned} H(e^{j\omega}) &= \sum_{r=0}^{L-1/2} h[L - r - \tfrac{1}{2}] (e^{-j(L-r-1/2)\omega} + e^{-j(L+r+1/2)\omega}) \\ &= e^{-jL\omega} \sum_{r=0}^{L-1/2} 2h[L - r - \tfrac{1}{2}] \cos((r + \tfrac{1}{2})\omega). \end{aligned} \quad (1.15)$$

All indices of h are integers, even though L itself is a half-integer. Again $\beta = 0$, but the cosine arguments now involve half-integer offsets.

Example. Take $(h[0], \dots, h[3]) = (\frac{1}{2}, 1, 1, \frac{1}{2})$, so $L = \frac{3}{2}$.

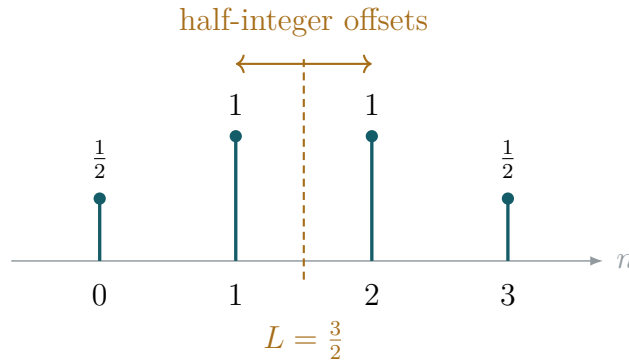


Figure 1.10: The closest samples are half a sample from the center, not one sample. This changes integer cosines into half-integer cosines.

The four terms can be grouped as

$$\begin{aligned} H(e^{j\omega}) &= \tfrac{1}{2} + e^{-j\omega} + e^{-2j\omega} + \tfrac{1}{2}e^{-3j\omega} \\ &= e^{-3j\omega/2} [(e^{j\omega/2} + e^{-j\omega/2}) + \tfrac{1}{2}(e^{3j\omega/2} + e^{-3j\omega/2})] \\ &= e^{-3j\omega/2} [2\cos(\omega/2) + \cos(3\omega/2)]. \end{aligned}$$

Thus the coefficients in $H_{\text{ampl}}(\omega) = \sum_{r=0}^{L-1/2} b[r] \cos((r + \frac{1}{2})\omega)$ are

$$b[r] = 2h[L - r - \tfrac{1}{2}], \quad 0 \leq r \leq L - \tfrac{1}{2}.$$

The moving average is the shortest Type II example: $L = \frac{1}{2}$ and $b[0] = 1$.

1.11 Antisymmetry: the center sample matters

For an antisymmetric filter, reflection changes the sign:

$$h[n] = -h[2L - n].$$

When L is an integer, substituting $n = L$ gives

$$h[L] = -h[L] \implies \boxed{h[L] = 0.}$$

It is not enough that the coefficients *away from* the center have opposite signs. The center has to satisfy the same reflection rule.

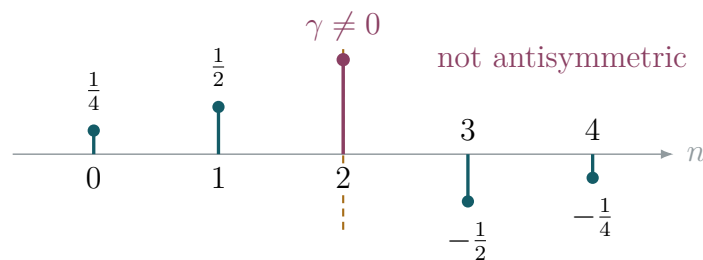


Figure 1.11: The nonzero center coefficient breaks antisymmetry, even though the outer pairs have opposite values. The symbol γ denotes the nonzero center height.

1.12 Type III: odd length, antisymmetric

For integer L , the center term vanishes and the reflected coefficients have opposite signs. Pairing them yields

$$\begin{aligned} H(e^{j\omega}) &= \sum_{r=1}^L h[L-r] (e^{-j(L-r)\omega} - e^{-j(L+r)\omega}) \\ &= e^{-jL\omega} \sum_{r=1}^L h[L-r] (e^{jr\omega} - e^{-jr\omega}) \\ &= e^{j(-L\omega+\pi/2)} \sum_{r=1}^L 2h[L-r] \sin(r\omega). \end{aligned} \quad (1.16)$$

We used $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$ and $j = e^{j\pi/2}$. Thus antisymmetry produces a sine amplitude and a constant phase offset $\beta = \pi/2$.

Type III example. Take

$$(h[0], \dots, h[4]) = (\tfrac{1}{2}, 1, 0, -1, -\tfrac{1}{2}), \quad L = 2.$$

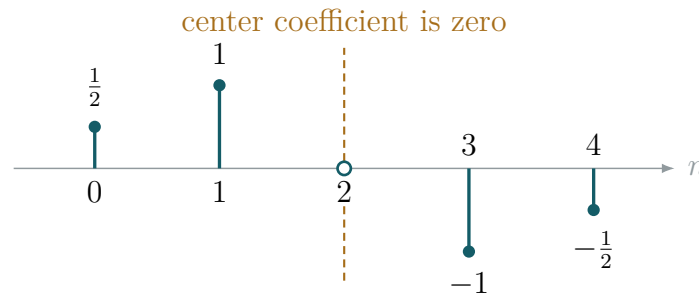


Figure 1.12: Opposite coefficients at integer offsets from the center. The nearest pair gives $2 \sin \omega$; the outer pair gives $\sin(2\omega)$.

Expanding and then pairing gives

$$\begin{aligned} H(e^{j\omega}) &= \tfrac{1}{2} + e^{-j\omega} - e^{-3j\omega} - \tfrac{1}{2}e^{-4j\omega} \\ &= e^{-2j\omega} [(e^{j\omega} - e^{-j\omega}) + \tfrac{1}{2}(e^{2j\omega} - e^{-2j\omega})] \\ &= e^{-2j\omega} j [2 \sin \omega + \sin(2\omega)] \\ &= e^{j(-2\omega+\pi/2)} [2 \sin \omega + \sin(2\omega)]. \end{aligned}$$

Writing the general amplitude as $\sum_{r=0}^L c[r] \sin(r\omega)$, we may take

$$c[0] = 0, \quad c[r] = 2h[L - r] \quad (1 \leq r \leq L).$$

The choice of $c[0]$ has no effect because $\sin(0\omega) = 0$.

The extra factor j is not a frequency-dependent distortion. It contributes the constant phase $\pi/2$. The slope of the phase factor is still $-L$, just as in the symmetric case. The distinction is the even or odd structure left after we remove the center exponential: cosine for symmetry, sine for antisymmetry.

1.13 Type IV: even length, antisymmetric

Here $L - \frac{1}{2}$ is an integer. There is no center sample to force to zero. Each pair is at offsets $\pm(r + \frac{1}{2})$, and its coefficients have opposite signs:

$$\begin{aligned}
 H(e^{j\omega}) &= \sum_{r=0}^{L-1/2} h[L - r - \tfrac{1}{2}] (e^{-j(L-r-1/2)\omega} - e^{-j(L+r+1/2)\omega}) \\
 &= e^{-jL\omega} \sum_{r=0}^{L-1/2} h[L - r - \tfrac{1}{2}] (e^{j(r+1/2)\omega} - e^{-j(r+1/2)\omega}) \\
 &= e^{j(-L\omega+\pi/2)} \sum_{r=0}^{L-1/2} 2h[L - r - \tfrac{1}{2}] \sin((r + \tfrac{1}{2})\omega). \tag{1.17}
 \end{aligned}$$

Thus the amplitude uses half-integer sine arguments and $\beta = \pi/2$.

Example. For $(h[0], \dots, h[3]) = (\frac{1}{2}, 1, -1, -\frac{1}{2})$, the center is $L = \frac{3}{2}$.

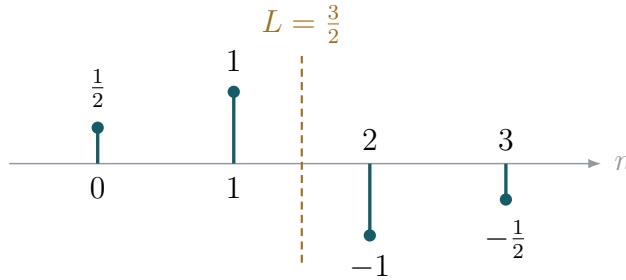


Figure 1.13: Opposite coefficients at half-integer offsets. There is no unpaired sample at the center.

Factoring the center exponential gives

$$\begin{aligned}
 H(e^{j\omega}) &= \tfrac{1}{2} + e^{-j\omega} - e^{-2j\omega} - \tfrac{1}{2}e^{-3j\omega} \\
 &= e^{-3j\omega/2} [(e^{j\omega/2} - e^{-j\omega/2}) + \tfrac{1}{2}(e^{3j\omega/2} - e^{-3j\omega/2})] \\
 &= e^{j(-3\omega/2+\pi/2)} [2 \sin(\omega/2) + \sin(3\omega/2)].
 \end{aligned}$$

In the general amplitude $\sum_{r=0}^{L-1/2} d[r] \sin((r + \frac{1}{2})\omega)$,

$$d[r] = 2h[L - r - \tfrac{1}{2}], \quad 0 \leq r \leq L - \tfrac{1}{2}.$$

1.14 Why these four cases give the characterization

The pair calculations prove that each type has linear phase. Conversely, suppose a real FIR filter has nonzero endpoints at 0 and $N - 1$ and admits

$$H(e^{j\omega}) = e^{j(-L\omega+\beta)} H_{\text{ampl}}(\omega)$$

with real H_{ampl} . We will recover both the center and the symmetry.

Conjugate this factorization and eliminate the real amplitude:

$$H(e^{j\omega}) = e^{-2jL\omega+2j\beta} \overline{H(e^{j\omega})}. \quad (1.18)$$

Since H is 2π -periodic and not identically zero, replacing ω by $\omega + 2\pi$ forces $e^{-4\pi jL} = 1$. Thus $2L$ is an integer. Extend h by zero outside its support and reindex the finite sum:

$$\begin{aligned} H(e^{j\omega}) &= e^{2j\beta} \sum_{n \in \mathbb{Z}} h[n] e^{-j(2L-n)\omega} \\ &= e^{2j\beta} \sum_{m \in \mathbb{Z}} h[2L - m] e^{-jm\omega}. \end{aligned}$$

To compare this with $\sum_m h[m] e^{-jm\omega}$, multiply by $e^{jr\omega}$ and integrate over $[-\pi, \pi]$. For integer q ,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{jq\omega} d\omega = \begin{cases} 1, & q = 0, \\ \frac{e^{jq\pi} - e^{-jq\pi}}{2\pi jq} = 0, & q \neq 0. \end{cases}$$

Only the term with $m = r$ survives. Therefore

$$h[r] = e^{2j\beta} h[2L - r] \quad (r \in \mathbb{Z}). \quad (1.19)$$

Since $h[0] \neq 0$, we have $h[2L] \neq 0$, hence $2L \leq N - 1$. Since $h[N - 1] \neq 0$, we have $h[2L - (N - 1)] \neq 0$, hence $2L \geq N - 1$. Consequently $L = (N - 1)/2$.

The two nonzero endpoint values in (1.19) are real. Thus $e^{2j\beta}$ is real and has modulus one: it is 1 or -1 . These give symmetry and antisymmetry, respectively. We may choose $\beta = 0$ or $\pi/2$, absorbing any remaining minus sign into H_{ampl} . Finally, the length is odd or even, giving exactly Types I–IV. The assumptions of a *real, nonzero FIR filter* are essential to this statement.

1.15 The four types at a glance

In every case,

$$H(e^{j\omega}) = e^{j(-L\omega+\beta)} H_{\text{ampl}}(\omega), \quad L = \frac{N-1}{2}.$$

The center controls the phase slope. The sign under reflection controls whether the amplitude is a cosine sum or a sine sum. The parity of the length controls whether the offsets are integers or half-integers.

TYPE I odd length, symmetric $H_{\text{ampl}} = \sum_{r=0}^L a[r] \cos(r\omega)$ $L \in \mathbb{Z}, \quad \beta = 0$	TYPE II even length, symmetric $H_{\text{ampl}} = \sum_{r=0}^{L-1/2} b[r] \cos((r + \frac{1}{2})\omega)$ $L - \frac{1}{2} \in \mathbb{Z}, \quad \beta = 0$
TYPE III odd length, antisymmetric $H_{\text{ampl}} = \sum_{r=1}^L c[r] \sin(r\omega)$ $L \in \mathbb{Z}, \quad \beta = \frac{\pi}{2}, \quad h[L] = 0$	TYPE IV even length, antisymmetric $H_{\text{ampl}} = \sum_{r=0}^{L-1/2} d[r] \sin((r + \frac{1}{2})\omega)$ $L - \frac{1}{2} \in \mathbb{Z}, \quad \beta = \frac{\pi}{2}$

Figure 1.14: Two choices give four types. The formulas were obtained by pairing coefficients at equal distances from the center, not by memorizing four unrelated identities.

The amplitudes in the examples were not normalized to have unit gain at zero frequency. For instance, the Type I example has $H(1) = 1 + 1 + \frac{1}{2} = \frac{5}{2}$. Multiplying all its coefficients by $2/5$ would normalize this gain without changing its type or its center. Symmetry and gain normalization are separate properties.

Likewise, the type alone does not say whether a filter is lowpass, highpass, or bandpass. The coefficients determine the amplitude response. But the type can force zeros at particular frequencies, which may rule out a proposed use. The final question gives an example.

1.16 Can a highpass filter be Type II?

Consider a highpass specification whose passband includes the Nyquist frequency $\omega = \pi$, so that the required gain there is nonzero. A Type II filter cannot satisfy this specification.

Indeed, every term in its real amplitude vanishes at $\omega = \pi$:

$$\cos\left((r + \tfrac{1}{2})\pi\right) = \cos(r\pi + \pi/2) = 0 \quad (r \in \mathbb{Z}).$$

Substitute this in (1.15):

$$\begin{aligned} H(e^{j\pi}) &= e^{-jL\pi} \sum_{r=0}^{L-1/2} b[r] \cos\left((r + \tfrac{1}{2})\pi\right) \\ &= 0. \end{aligned}$$

Thus a zero at π is forced by the combination of even length and symmetry, independently of the values of $b[r]$.

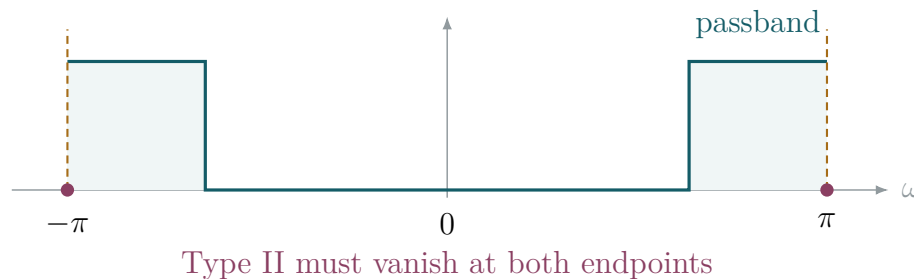


Figure 1.15: A desired highpass magnitude with nonzero endpoint gain. The filled dots mark the zeros that every Type II filter must have. The rectangular response is a specification sketch, not a claim of an exact FIR realization.

What to remember

Linearity and time invariance turn an input into a superposition of shifted impulse responses: $y = h * x$. Complex exponentials turn that convolution into a scalar multiplication. For real FIR filters, pairing coefficients about their center exposes linear phase: equal pairs give cosines, and opposite pairs give sines.

This is why time-domain structure is useful in filter design. A condition on a few coefficients can impose an identity at every frequency. Many wavelet filters have linear phase. When we encounter them, these calculations will let us recognize the symmetry and understand the restrictions it imposes on their frequency responses.